

John Abrahams

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Software engineer and M.S.E. Data Science candidate (UPenn, Dec 2026) who builds production ML and low-latency systems end-to-end, across Rust, Python, Java, and C++.

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science

December 2026

M.S.E. in Data Science · GPA 3.62

Relevant coursework: Probability Theory (Ph.D. course), Advanced Topics in Operating Systems, Advanced Topics in Computer Architecture, Big Data Analytics, Theoretical & Computational Neuroscience, Smart Devices

Lehigh University, P.C. Rossin College of Engineering and Applied Science

May 2025

B.S. Computer Science & Engineering · GPA 3.65 (Major 3.75) · Minor: Probability & Statistics · Dean's List Spring 2023 – Spring 2025

Relevant coursework: Applied Machine Learning, Blockchain and Database Systems

EXPERIENCE

New York Brain & Spine Surgery, Software Engineer

May 2025 – March 2026

- Built a production therapy chatbot on LLaMA and FastAPI, blending rule-based clinical workflows with LLM-generated responses.
- Designed a configurable multi-agent architecture supporting multiple mental-health conditions and rapid prompt iteration.
- Developed backend APIs and state management powering safe, real-time conversational flows and fast end-to-end iteration.

Ninth Wave, Software Engineer Intern, Compass (open-finance AI onboarding)

May 2026 – Aug 2026

- Researching a two-model code-generation system for Compass, Ninth Wave's open-finance onboarding agent: an HRM-Text recursive executive that plans and writes structured outputs over a MeMo parametric memory trained on internal documentation, replacing RAG and prompt engineering with learned memory.
- Contributed to Flight-Deck (Java/Spring Boot developer portal) and the Compass agent on Amazon Bedrock AgentCore, working across AWS (Secrets Manager, OpenSearch, Bedrock).
- Built Python LLM tooling for bank-API-to-canonical-model mapping and automated RFP response generation across ERP integrations (Xero, NetSuite, QuickBooks).

Kronos: Multi-Tenant WebAssembly Runtime, UPenn Research (Prof. Sebastian Angel)

Oct 2024 – Present

- Built Kronos, a multi-tenant WebAssembly runtime in Rust (~5K lines) that packs tenants ~20× denser than Docker (~0.08 MiB per isolate) using exokernel-style capability gating with per-invocation fuel and wall-clock budgets.
- Engineered a work-stealing thread pool that dispatches guest pthread_create calls for shared-memory data parallelism, reaching 6.7× speedup on four cores.
- Exposed a tiered ~74-entry capability-gated host interface; benchmarked against the wasmtime and wazero runtimes on C++ kernels (matmul, k-means, SVD) and co-authored a 9-page systems paper introducing the resulting 'Type II exovisor' abstraction.

The Quant Game, Co-Founder

May 2024 – Aug 2024

- Co-founded and built a full-stack options-strategy simulator (React, Python) delivering real-time backtests and performance metrics.
- Architected an AWS data layer storing and serving 61M+ financial data points at high availability.

Systematic Trading Bot, Independent

Apr 2026 – Present

- Built a systematic daily-rebalance U.S. ETF momentum strategy on the Alpaca API, validated through a vectorbt backtesting harness.
- Automated live paper trading on AWS EC2 with Secrets Manager and systemd timers for unattended daily execution.

Capstone Document Search System, Senior Capstone, Lehigh University

May 2024 – July 2025

- Engineered a full-stack search engine over 23,000+ Dropbox documents on AWS for fast, sentence-level content retrieval.
- Built an NLP indexing pipeline that parses heterogeneous file types and segments content at the sentence level for granular highlight extraction.

SKILLS

Languages: Python, Rust, Go, Java, C++, C, WebAssembly

Machine Learning: PyTorch, Reinforcement Learning, GRU/sequence models, LLMs, NLP

Systems: AWS, FastAPI, Spring Boot, embedded/ESP32, multithreading, concurrency

Data: probability & statistics, regression, SVM, EM algorithm